

# Donavan Borges

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## EDUCATION

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### • California Institute of Technology (Caltech)

Expected December, 2026

B.S. in Computer Science (Robotics Track)

Pasadena, CA

- **Relevant Coursework:** Experimental Robotics; Robotics (a & b); Learning Systems; Mechatronics; Signal Processing; Machine Learning; Algorithms

## EXPERIENCE & PROJECTS

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### • Autonomous Combat Robot Lead

Sep 2025 – Present

Caltech Robotics Team

- Leading Caltech's first autonomous combat robotics project, integrating YOLO vision models trained in Roboflow with an external Intel RealSense camera for onboard perception.
- Developing full autonomy stack: real-time object detection, tracking, embedded control, and decision-making; validating algorithms through Unity/Unreal Engine simulations.
- Building communication pipeline using an ESP-32 on the chassis to receive commands from a laptop-based control system.
- Collaborating with controls and hardware subteams to integrate autonomy into a combat-ready platform.

### • Autonomous Multi-Robot Systems

Mar 2025 – Jun 2025

ME/CS/EE 129: Experimental Robotics

- Programmed mobile robots for exploration, mapping, navigation, and reactive obstacle avoidance using SLAM and Dijkstra's Algorithm.
- Designed multi-threaded architectures with ROS threads running on a Raspberry Pi 4 using an Ubuntu virtual machine development environment.
- Implemented motor control through continuous sensor feedback loops (IR, ultrasound) using both polling and interrupt-driven I/O for real-time decision making.
- Built planning and decision modules with decision trees and graph-based exploration algorithms; delivered a multi-robot autonomous system for the final demonstration.

## SKILLS

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- **Programming & Simulation:** Python, C++, MATLAB, ROS2; linux; RVIZ/ISSAC SIM/Mujoco/UNREAL ENGINE/UNITY; multi-threaded programming
- **Robotics & Systems:** Motor control; sensor integration; feedback control; planning & navigation
- **Hardware Platforms:** Embedded systems, Arduino, Raspberry Pi; ESP32; autonomous mobile robots
- **Other:** Team leadership; project planning; fast iteration, adaptability, communication, quick-learning

## LEADERSHIP EXPERIENCE

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### • Treasurer & Business/Software Team Lead

Apr 2025 – Present

Caltech Robotics Team

- Managed finances for multi-project robotics initiatives; raised over \$15K in the first term toward an annual \$50K goal.
- Led software development across projects (open-source quadruped, combat robots, RoboSub, and team website), ensuring consistent workflows using Git and modern CI/CD practices.
- Onboarded and trained new members to help them ramp up quickly on robotics software stacks, ROS pipelines, and Git workflows.
- Coordinated outreach events including cross-university combat robotics competitions, lab tours, and STEM outreach programs.
- Established sponsorship relationships, drafted proposal documents, and collaborated with companies to secure equipment and funding.

## AWARDS

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### • Questbridge National College Match Scholarship

Dec 2020

Questbridge

- Full four-year scholarship awarded for outstanding academic achievement and leadership potential among high-achieving, low-income students nationwide